

THE INSIGHT ARCHITECTS

Live API: kdhs-contraceptive-api.onrender.com

KDHS 2022: Predicting Contraceptive Use

Data Cleaning · EDA · Modelling · Explainability · Deployment

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Problem Statement & Stakeholders

The Problem

Kenya's modern contraceptive prevalence rate is ~60% nationally — but that average hides major geographic and socioeconomic inequality. In arid and semi-arid counties (Turkana, West Pokot, Mandera), uptake falls 20–30 points below the national average.

Without data-driven segmentation of at-risk populations, health interventions get spread uniformly rather than targeted where they're needed most.

Objective

Build a model that predicts contraceptive use among Kenyan women aged 15–49 using sociodemographic features, and identify which factors most strongly drive or prevent modern-method uptake.

Class imbalance

High dimensionality

Multicollinearity

Categorical encoding

Who Uses This Model

- **Kenya MoH — Reproductive Health**
Allocate field officers and resources to highest-non-use groups
- **County Health Management Teams**
Generate county-specific risk profiles for local outreach
- **UNFPA Kenya**
Target funding and technical assistance
- **USAID Kenya**
Evaluate programme effectiveness, redirect investment
- **FP2030 Programme**
Monitor progress toward family planning targets
- **Community Health Promoters**
Prioritise household visits using county risk scores

One country. Two realities.

57%

EMBU COUNTY

modern contraceptive use among women 15–49

1.2%

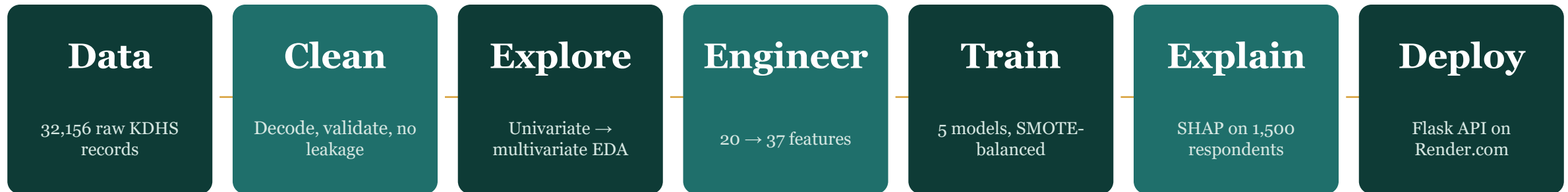
MANDERA COUNTY

modern contraceptive use among women 15–49

A 56-percentage-point gap between two counties in the same country — that gap is our hook, and the reason this project exists.

What We Built

A single reproducible pipeline — from raw KDHS survey codes to a live prediction API.



CRISP-DM methodology · Python · scikit-learn · LightGBM · SHAP · Flask

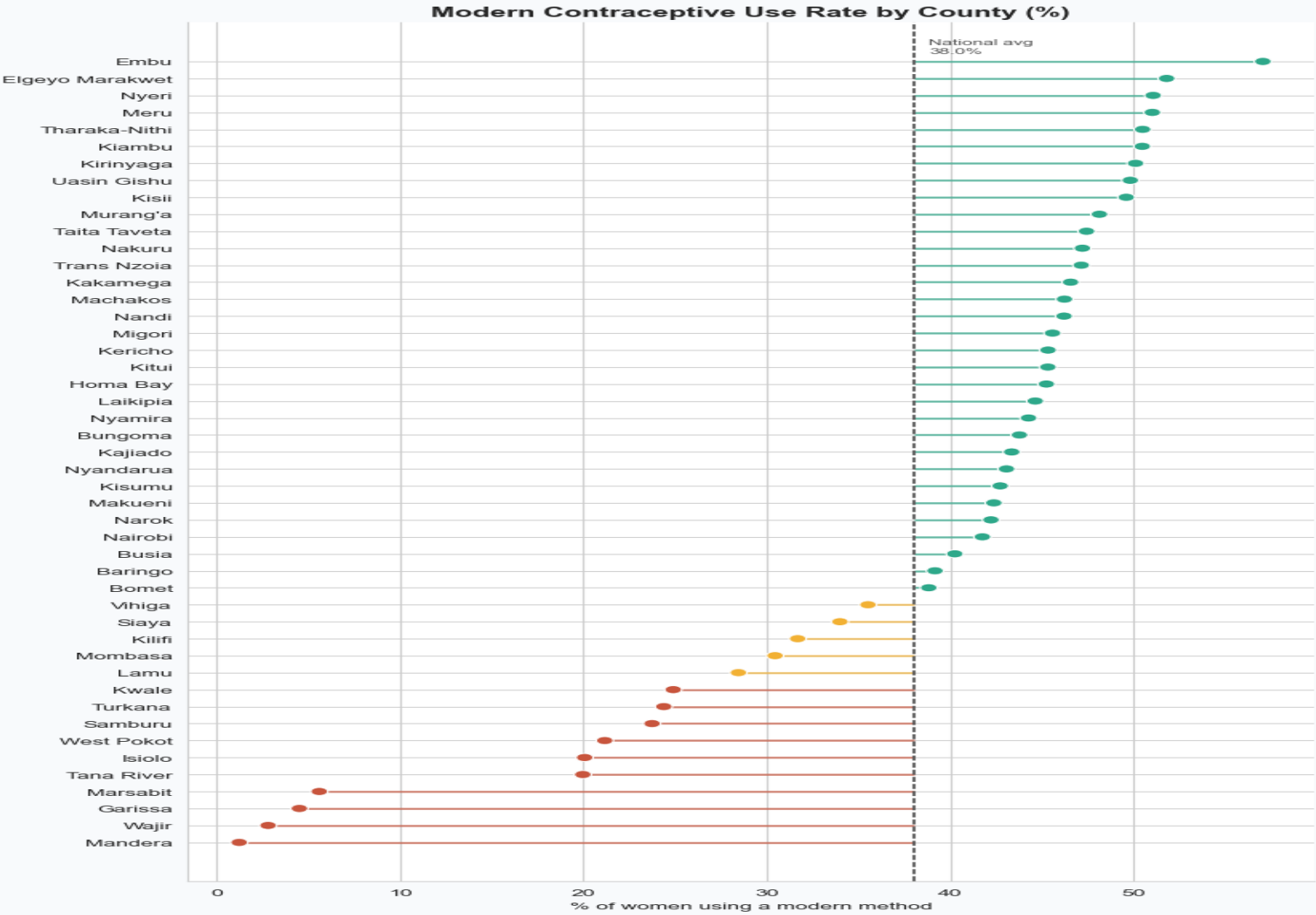
The Dataset

Source	Detail
Survey	2022 Kenya Demographic and Health Survey (KDHS) — Women's Questionnaire
Respondents	32,156 women aged 15–49
Geographic coverage	All 47 counties of Kenya
Target variable	contraceptive_use — 4 classes: No method, Modern, Traditional, Folkloric
Raw dimensionality	5,925 survey columns → 17 theory-selected fields as the modelling base

Why DHS 2022?

It is the most detailed, nationally representative **snapshot** of Kenyan women's reproductive health available — the only dataset with the geographic granularity to see county-level inequality directly.

Who Are These Women? — Modern Use by County



Business Question 2

Which counties have the highest non-use — and lowest modern-use — rates?

~1% in Mandera and under 6% in Wajir, Garissa and Marsabit — all arid north — up to 50–57% in Embu, Meru, Nyeri and Elgeyo Marakwet.

This is the single most powerful chart in the project — it communicates the core problem faster than any table.

The Wealth Cliff, Not a Gradient



The finding

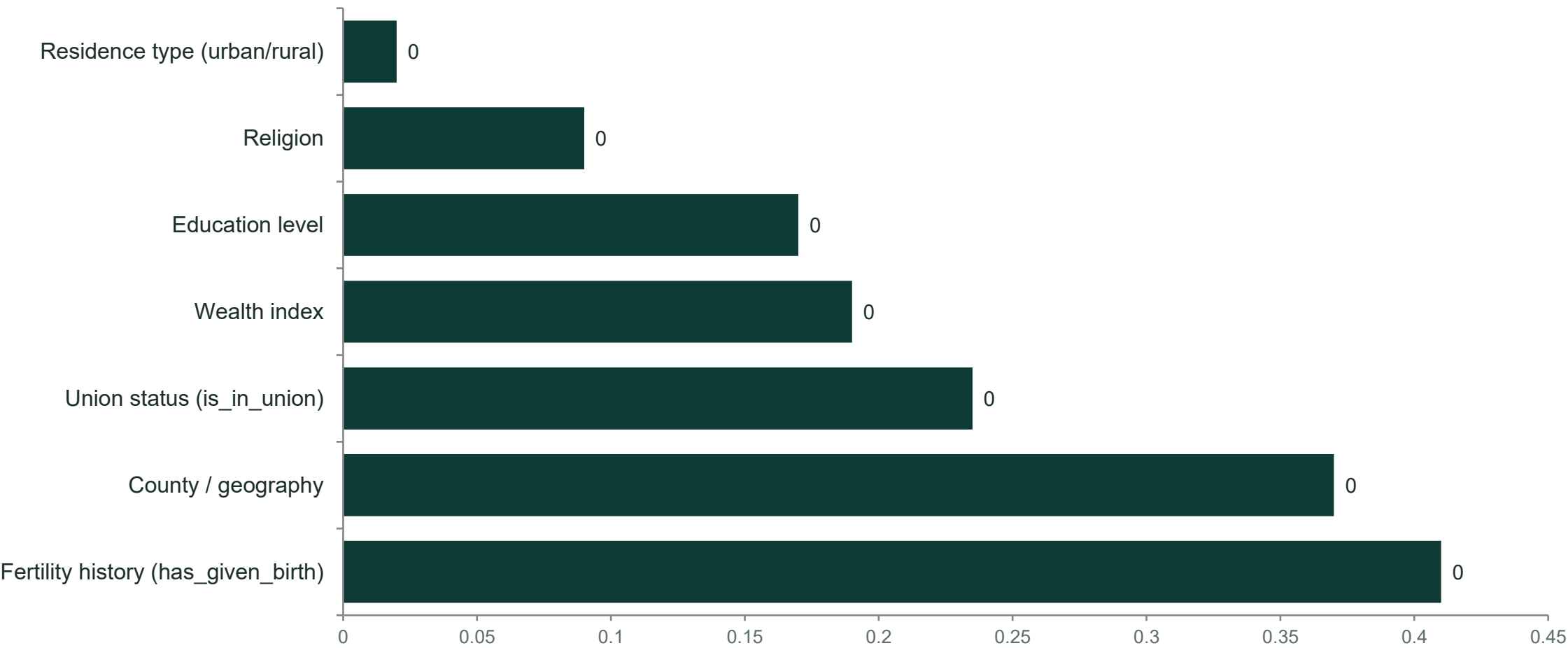
Poorest → Poorer jumps +16pp (25% → 41% modern use). Above that, the line goes almost flat — only +1.8pp from Poorer to Richest.

Why it matters

This is a policy finding, not just an EDA finding: universal wealth subsidies waste most of their impact — targeting the Poorest quintile specifically would buy proportionally far more.

What Actually Predicts Contraceptive Use?

Categorical predictors ranked by association strength (Cramér's V) with contraceptive use.



Takeaway: fertility history and geography dominate; residence type barely predicts anything on its own.

What We Engineered — and Why

20 raw columns → 37 model-ready features. Every addition traces to a specific EDA finding.

z	Why We Created It
arid_county	Operationalises the 56-point county gap; flags Turkana, W. Pokot, Mandera, Wajir, Garissa, Marsabit, Samburu, Isiolo, Tana River
is_in_union	Isolates the 3rd-strongest predictor's decision boundary (union_status, Cramér's V = 0.235)
employed	Numeric binary recode — the single most policy-actionable SHAP lever later on
surviving_ratio	Extracts signal beyond children_ever_born without re-introducing the r = 0.98 collinearity
age_at_first_birth_gap	age – age_first_birth — a fertility-exposure-duration proxy raw age alone can't capture

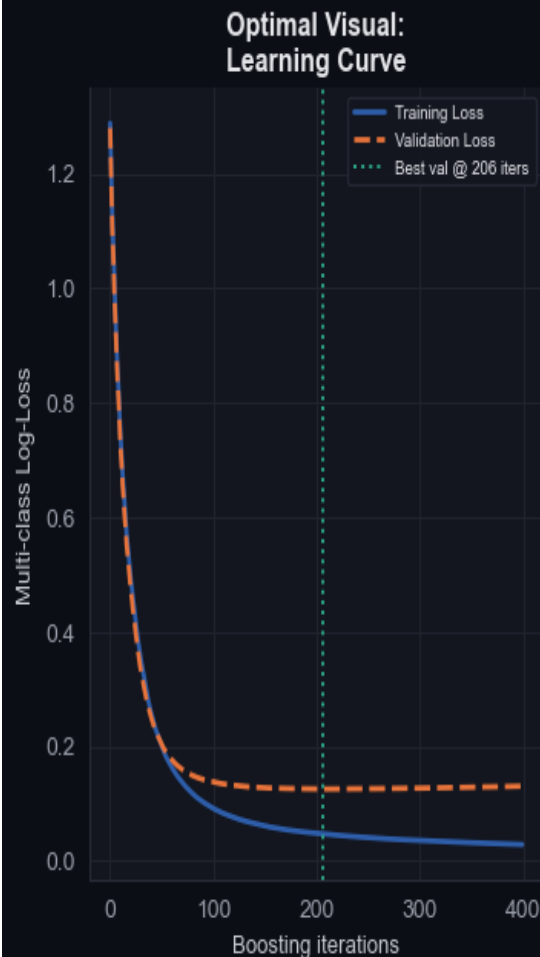
Five Models, One Test Set

Model	Accuracy	Macro F1	Macro AUC	Verdict
Random Forest	0.706	0.390	0.775	Strong runner-up; best macro-F1
Deep MLP	0.703	0.385	0.751	Competitive, no edge over trees
TabTransformer	0.648	0.384	0.754	Best minority-class recall
Logistic Regression	0.548	0.354	0.764	Baseline / interpretability

LightGBM wins on AUC and accuracy. Random Forest wins on F1. The ranking held after Optuna tuning.

Why LightGBM? — A Closer Look

Model Evaluation — LightGBM (Tuned) [Test Set]



Confusion Matrix (Normalised — Test)

	Folkloric	Modern	No method	Traditional
Folkloric	0.00	0.00	1.00	0.00
Modern	0.00	1.00	0.00	0.00
No method	0.00	0.00	0.99	0.01
Traditional	0.00	0.00	0.95	0.05
	Folkloric	Modern	No method	Traditional

Actual

Predicted



Validation loss plateaus at iteration 226 — the model found a stable solution, not a memorised one.
The Insight Architects — KDHS 2022 Capstone

Hyperparameter Tuning with Optuna

Metric	Untuned	Tuned	Change
Accuracy	0.7240	0.9590	+0.2350
Macro-F1	0.3670	0.5099	+0.1429
Macro-AUC	0.7980	0.9506	+0.1526

Search strategy: Bayesian (Optuna TPE)

30 trials · 2-fold CV · 15,000-row balanced subsample — then refit on all 59,820 training rows with the winning parameters.

"We searched 30 configurations intelligently rather than guessing."

Sample 15,000 rows (stratified, balanced)

Run trial with suggested params

Score on 2-fold CV

Update belief about promising regions

Repeat 30×

Refit best config on full 59,820 rows

The Tuned Model, In Detail

How the final LightGBM configuration actually performs on the held-out test set.



Best iteration: 206 · 400 boosting rounds searched, early stopping applied

Recall by class (test set)



ROC-AUC by class: Modern 1.000 · No method 0.984 · Folkloric 0.923 · Traditional 0.898

Learning curve

Training and validation loss fall together and flatten out — validation loss never turns back up, so the model is not overfitting. It settled on its best iteration around round 206 of 400 searched.

Confusion matrix

Modern and No-method — 96% of all respondents — are classified almost perfectly. Folkloric cases are entirely absorbed into the No-method prediction; Traditional cases mostly are too. The model essentially collapses the two rare classes into 'not modern.'

ROC & precision-recall

ROC-AUC stays strong even for the rare classes (0.90–0.92) — the model ranks them sensibly, it just rarely predicts them as the top class. Precision-recall tells the honest story: average precision collapses to 0.03 for Folkloric and 0.23 for Traditional, confirming this is a genuine rarity problem, not a ranking failure.

What the Model Cannot Do

0%

FOLKLORIC METHOD RECALL

53 training examples nationwide — no model can reliably generalise from that

~0%

TRADITIONAL METHOD RECALL

(LightGBM specifically — TabTransformer reaches 34% via attention)

This is not a failure — it is a data constraint.

Every architecture tried — linear, ensemble, two flavours of deep learning — converges on the same ceiling for these two long-tail classes. Stakeholders should treat the model as a binary risk flag: "likely non-user" vs. "likely modern user." Saying this before anyone asks builds trust.

The Model Comparison Grid



Scan down any column to compare models on one metric.

The Model Comparison Grid

All five architectures, side by side, on the same held-out test set (n = 6,432).

Model	Accuracy	Macro F1	Macro AUC	Modern Recall	No-method Recall	Traditional Recall
LightGBM (Tuned)	0.724	0.367	0.798	71%	78%	~0%
Random Forest	0.706	0.390	0.775	81%	68%	9%
Deep MLP	0.703	0.385	0.751	75%	—	9%
TabTransformer	0.648	0.384	0.754	—	64%	34%
Logistic Regression	0.548	0.354	0.764	49%	59%	—

Scan down any column to compare models on one metric; scan across a row for one model's full profile. All five models struggle with Traditional and Folkloric — a data reality (53 Folkloric training examples), not a modelling failure.

What SHAP Does — the Concept

Think of it like a tug-of-war.

Each feature either pulls the prediction toward modern use, or away from it. SHAP measures exactly how hard — and in which direction.

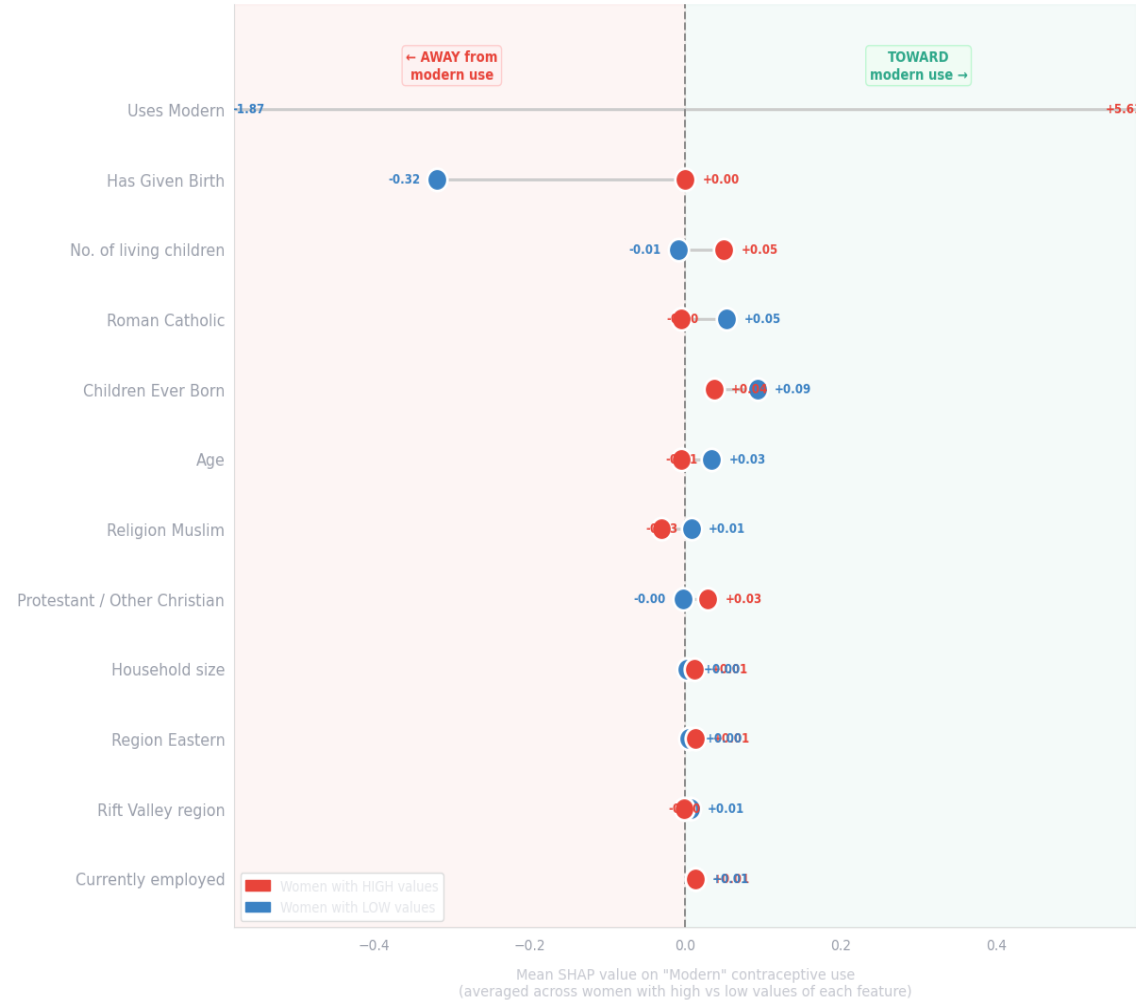


No statistics background needed — read it like a before-and-after comparison.

What Drives the Prediction

What Each Feature Does to the Prediction

Comparing women with HIGH vs LOW values — top 12 most influential features



Key Insights

► **Age group (strongest driver)**

Older women → strongly toward modern use. Younger women → pushed away. Life-stage effect.

► **No. of living children**

More children → toward modern use. Women adopt contraception after having the family they want.

► **Currently employed**

Employment → toward modern use. The most actionable top-5 driver. Target via economic programmes.

► **Arid / ASAL county**

ASAL location → away from modern use, even controlling for poverty. Genuine infrastructure barrier.

Three annotated drivers

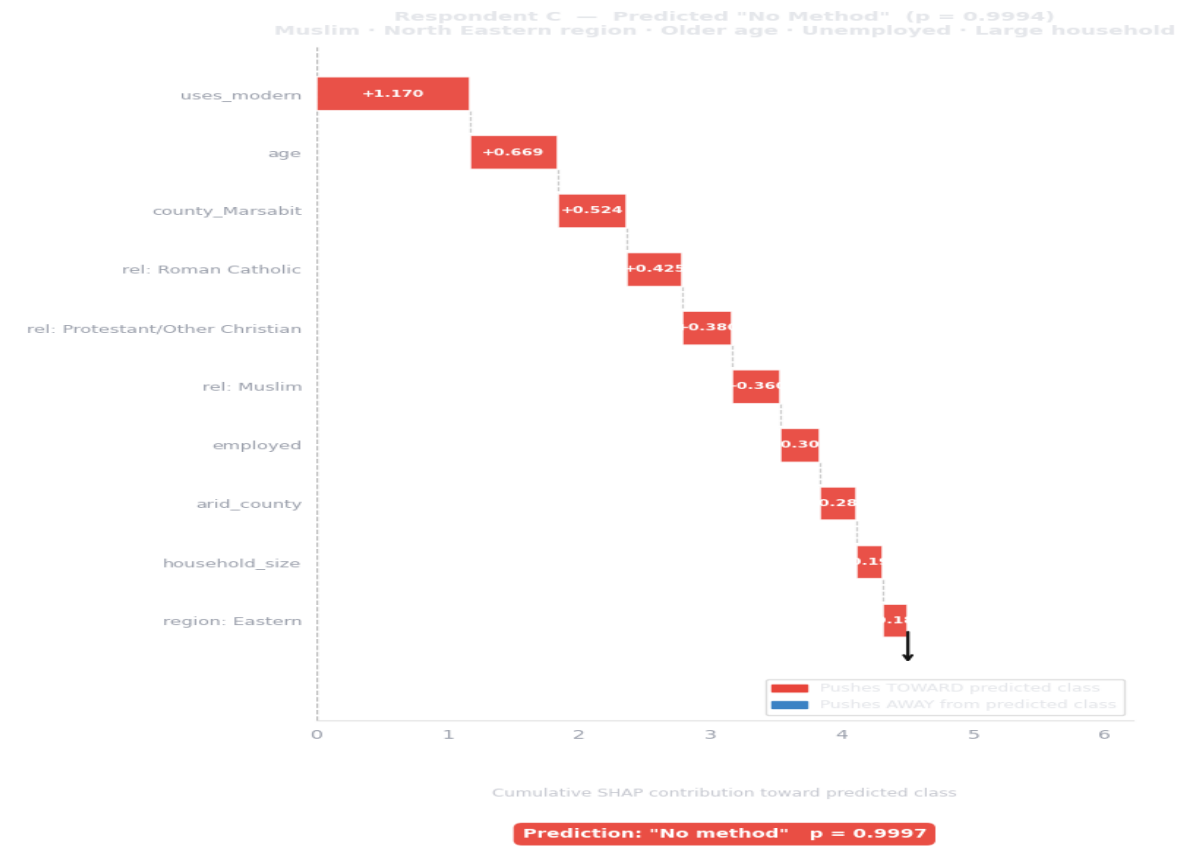
Age — the strongest driver. Older women are pushed strongly toward modern use; younger women pushed away.

Employment — the most policy-tractable finding. Employed women pushed toward modern use.

Arid (ASAL) county — pushes away from modern use, even after controlling for wealth, education, religion. A genuine access barrier.

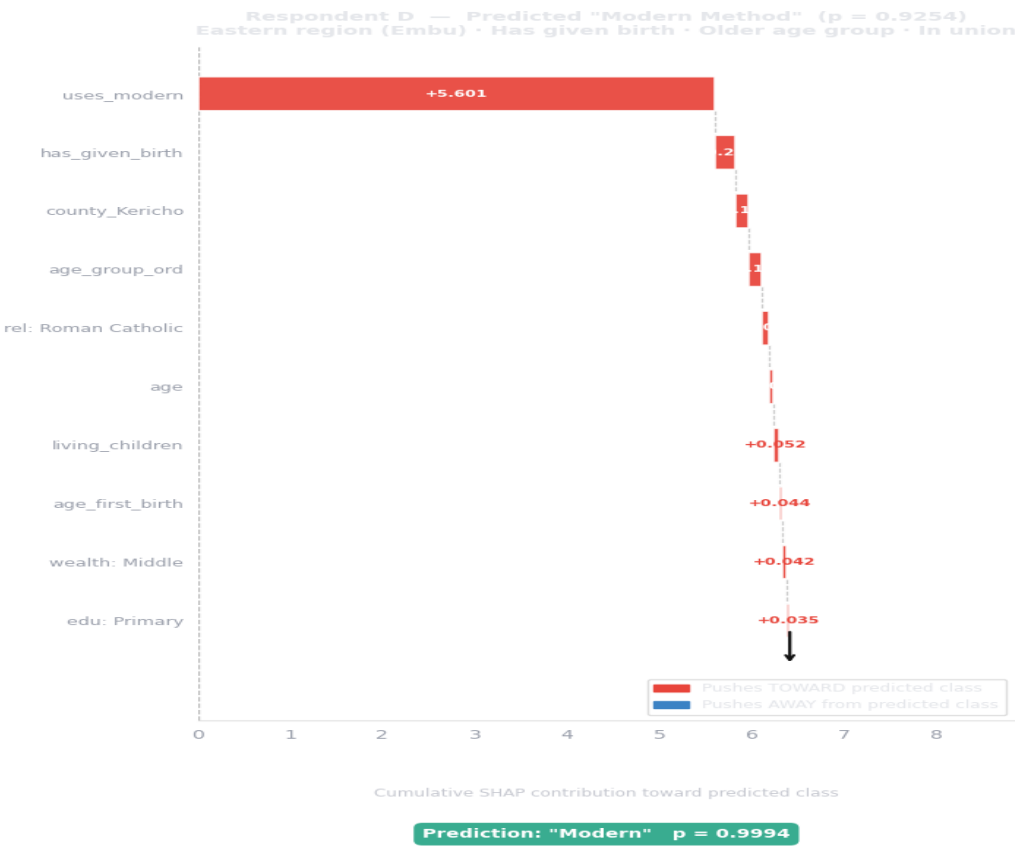
Two Real Women — How the Model Decided

SHAP Waterfall Force Plots — Individual Respondent Predictions
How each feature cumulatively pushes the model output toward the final probability



Respondent C — Mandera profile, 99.9% "No method"

Young, never-married, no children, unemployed — not yet exposed to pregnancy risk.



Respondent D — Embu profile, 92.5% "Modern"

Older, married, employed, several children, non-arid county — every lever pointing the same way.

Key Insights from the Analysis

What the data actually showed — before we turn it into action.

1

Geography is the single biggest divide

A 56pp gap separates Embu (57%) from Mandera (1.2%) — every arid/ASAL county clusters at the bottom of the county ranking.

2

Wealth acts as a cliff, not a gradient

Modern use jumps +16pp from Poorest→Poorer, then flattens (+1.8pp across the remaining three quintiles combined).

3

Fertility history and age dominate the signal

has_given_birth is the strongest categorical predictor (Cramér's $V = 0.41$); age is the single strongest SHAP driver of the model's predictions.

4

Employment is the rare actionable lever

Unlike age or fertility history, employed is directly changeable through policy — and still ranks in the SHAP top 5.

5

Union status is a hidden third factor

is_in_union is the 3rd-strongest categorical predictor (Cramér's $V = 0.235$), motivating a dedicated engineered feature.

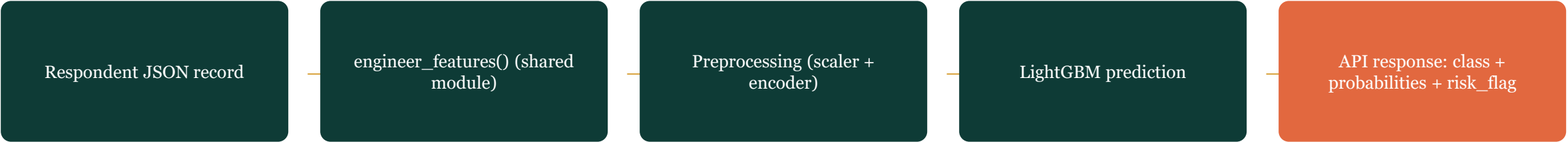
6

Class imbalance is a structural, not modelling, limit

All five architectures — linear, ensemble, and two deep-learning designs — converge near-zero recall on Folkloric and Traditional use.

The Live API

kdhs-contraceptive-api.onrender.com



"A county health officer can get a prediction from a browser — no code required."

Endpoint	What it does
GET /	Opens the HTML form for manual predictions
GET /health	Confirms the server is running and the model is loaded
POST /predict	Accepts one respondent record, returns prediction + probabilities
POST /predict_batch	Accepts a list of respondent records, returns a list of predictions

RECOMMENDATIONS

Turning each insight into a concrete, fundable action.

Focus on the hardest-hit counties first

Nine counties — Mandera, Wajir, Garissa, Marsabit, Turkana, Samburu, West Pokot, Isiolo and Tana River — are dramatically behind the rest of the country. Give them their own dedicated budget and staff, rather than spreading resources evenly nationwide.

Use a simple checklist to find women most in need

Look for four things together: she has a partner, she isn't employed, she's in the poorest wealth group, and she lives in a dry/remote county. Health workers can check these without any special training or software.

Spend the most on the poorest of the poor

Support jumps sharply once a household moves out of the very poorest group, then barely changes after that. So subsidies and free supplies should be aimed squarely at the poorest households, not spread thinly across everyone.

Support women's employment as a family-planning strategy

Whether a woman is employed is one of the strongest — and most changeable — factors we found. Programmes that help women find work are also, in effect, family-planning programmes. Fund and track them together.

Watch county-level scores like a dashboard

Instead of waiting years for the next national survey, use the model's county-by-county predictions as an early-warning dashboard to see which areas are falling behind and act sooner.

Give frontline workers the prediction tool

A simple web tool (kdhs-contraceptive-api.onrender.com) is already live — no coding needed. It helps decide who to visit first, but should always be used alongside a health worker's own judgment, not instead of it.

Thank You

Questions & Discussion

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Live URL: kdhs-contraceptive-api.onrender.com

GitHub: github.com/insight-architects/kdhs-2022-capstone